

# A “Route 2” proof of [Lemma 88](#): the tame fundamental group of $U^{(d)}$ and the compactified Abel–Jacobi fibration

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**Goal.** The proof of [Theorem 2](#) (“Theorem 2”) reduces, in the ramified case  $\mathfrak{m} > 0$ , to the extension lemma whose proof in the thesis is currently a placeholder (`ProofOfReducedTheorem.tex`): a locally constant sheaf  $\mathcal{F}^{[d]}$  on  $U^{(d)}$  which is *tamely* ramified along the boundary divisor  $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$  extends to a locally constant sheaf on all of  $\tilde{C}_{\mathfrak{m}}^{(d)}$ . The placeholder suggests two routes; this note carries out **Route 2**, made precise as follows: for every finite abelian  $G$ , pullback along  $\Phi_d$  identifies characters of  $\pi_1^{et}(\mathrm{Pic}_{C,\mathfrak{m}}^d)$  with the *tame* characters of  $\pi_1^{et}(U^{(d)})$  — so the tame abelianized fundamental group of  $U^{(d)}$  maps *isomorphically* (not merely injectively) onto  $\pi_1^{et,ab}(\mathrm{Pic}_{C,\mathfrak{m}}^d)$ . Route 1 (the kernel of  $\pi_1^{ab}(U^{(d)}) \rightarrow \pi_1^{ab}(\mathrm{Pic}_{C,\mathfrak{m}}^d)$  is pro- $p$ ) is deduced at the end as a corollary.

The proof proceeds in two independent steps: by Zariski–Nagata purity, it suffices to show the corresponding torsor is *unramified* at the generic point  $\eta_0$  of  $H$ ; then tameness at  $\eta_0$  is upgraded to unramifiedness. For the second step, instead of computing valuations on the blowup of  $C^{(d)}$ , we work in the *generic fiber*  $P_\xi \cong \mathbb{P}_\kappa^N$  of the compactified Abel–Jacobi morphism  $\tilde{\Phi}_d$ : the point  $\eta_0$  lies in  $P_\xi$  with the same local ring, the boundary meets  $P_\xi$  in a divisor which is geometrically a hyperplane (its degree over  $\kappa$  is a power of  $p$ ), and the degree-zero property of principal divisors on projective space then forces the tame inertia at  $\eta_0$  to vanish.

All numbered references to results *not* proved in this note (Takeuchi’s compactification, the degree condition, the tameness definitions, purity inputs, etc.) carry the numbering of the compiled thesis (`main.pdf`).

## The extension lemma, via the compactified Abel–Jacobi fibration (“Route 2”)

We keep the notation of [Section 4.4](#):  $C$  is a projective smooth geometrically connected curve over the perfect field  $k$ ,  $\mathfrak{m} = \sum n_i P_i > 0$  is a modulus with complement  $U = C \setminus \mathfrak{m}$ , and  $d$  is an integer satisfying [\(10\)](#). We write  $Z_0 = Z_0(\mathfrak{m}, d) \subset C^{(d)}$  for the center of the blowup  $\pi : X_{\mathfrak{m},d} \rightarrow C^{(d)}$ ,  $E_0$  for its exceptional divisor with generic point  $\eta_0$ ,  $\tilde{C}_{\mathfrak{m}}^{(d)} \subseteq X_{\mathfrak{m},d}$  for Takeuchi’s compactification ([Theorem 80](#)), and  $H = \tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)} = E_0 \times_{C^{(d)}} \tilde{C}_{\mathfrak{m}}^{(d)}$  for the boundary divisor. Throughout,  $p$  denotes the characteristic of  $k$ ; if  $p = 0$ , statements of the form “prime to  $p$ ” are to be read as no condition, and the arguments below only simplify.

**Lemma 1.** *If  $\mathcal{F}^{[d]}$  is a locally constant sheaf on  $U^{(d)}$  which is tamely ramified along the boundary divisor  $H = \tilde{C}_m^{(d)} \setminus U^{(d)}$ , then  $\mathcal{F}^{[d]}$  extends to a locally constant sheaf  $\tilde{\mathcal{F}}^{[d]}$  on  $\tilde{C}_m^{(d)}$ .*

The proof proceeds in two steps. First (Proposition 3), by the Zariski–Nagata purity of the branch locus, extending  $\mathcal{F}^{[d]}$  across  $H$  is possible as soon as the corresponding torsor is *unramified* at the generic point  $\eta_0$  of  $H$ ; a priori tameness is weaker than this. Second (Proposition 7), we show that at  $\eta_0$  tameness in fact forces unramifiedness. This is where the global geometry enters, and where “Route 2” differs from a direct computation on the blowup: the point  $\eta_0$  is itself a point of the generic fiber  $P_\xi \cong \mathbb{P}_\kappa^N$  of the compactified Abel–Jacobi morphism  $\tilde{\Phi}_d : \tilde{C}_m^{(d)} \rightarrow \text{Pic}_{C,m}^d$ , the boundary  $H$  meets  $P_\xi$  in an irreducible divisor whose complement is (geometrically) an affine space, and hence that divisor is geometrically a *hyperplane*. Since a principal divisor on projective space has degree zero, a rational function whose divisor is, modulo  $m$ , supported on a single divisor of degree prime to  $m$  must have  $v_{\eta_0}$ -valuation divisible by  $m$ ; applied to a Kummer representative of (the prime-to- $p$  part of) our torsor, this kills the tame inertia at  $\eta_0$ . At the end we record the resulting description of the tame abelianized fundamental group of  $U^{(d)}$  (Corollary 8), which is the precise form of “Route 2”: the tame abelianized fundamental group of  $U^{(d)}$  maps isomorphically (not merely injectively) onto  $\pi_1^{et,ab}(\text{Pic}_{C,m}^d)$ .

We first record the relevant geometry of the pair  $(\tilde{C}_m^{(d)}, H)$ .

**Lemma 2.** *With notation as above:*

1.  $\tilde{C}_m^{(d)}$  is an integral scheme, smooth over  $k$ .
2.  $H$  is a nonempty open subscheme of  $E_0$ . In particular  $H$  is irreducible of codimension 1 in  $\tilde{C}_m^{(d)}$ , its generic point is  $\eta_0$ , and  $\eta_0 \in \tilde{C}_m^{(d)}$ .
3.  $\eta_0$  is the unique point of  $\tilde{C}_m^{(d)} \setminus U^{(d)}$  whose local ring has dimension 1, and  $\mathcal{O}_{\tilde{C}_m^{(d)}, \eta_0} = \mathcal{O}_{X_{m,d}, \eta_0}$  is a discrete valuation ring with fraction field  $K := k(C^{(d)})$ .

*Proof.* (1)  $\tilde{C}_m^{(d)}$  is an open subscheme of  $X_{m,d}$ , which is integral as the blowup of the integral scheme  $C^{(d)}$ ; hence  $\tilde{C}_m^{(d)}$  is integral. By Theorem 80,  $\tilde{C}_m^{(d)}$  is a projective space bundle over  $\text{Pic}_{C,m}^d$ . The latter is smooth over  $k$ :  $\text{Pic}_{C,m}^0$  is a smooth group scheme and  $\text{Pic}_{C,m}^d$  is a torsor under it (see the properties of the generalized Picard scheme recalled in Section 4). A projective space bundle over a smooth  $k$ -scheme is smooth over  $k$ .

(2) Since  $\tilde{C}_m^{(d)} \hookrightarrow X_{m,d}$  is an open immersion, its base change  $H = E_0 \times_{C^{(d)}} \tilde{C}_m^{(d)} = E_0 \times_{X_{m,d}} \tilde{C}_m^{(d)} \hookrightarrow E_0$  is an open immersion as well. To see  $H \neq \emptyset$ , suppose otherwise; then  $U^{(d)} = \tilde{C}_m^{(d)}$  and  $\Phi_d = \tilde{\Phi}_d$  would be a projective space bundle. But the geometric fibers of  $\Phi_d$  are affine spaces of dimension  $d - \deg m - g + 1 \geq 1$  (Section 4.3; positivity follows from (10)), and a positive dimensional affine space over a field is not proper — a contradiction. A nonempty open subscheme of the irreducible scheme  $E_0$  is irreducible, dense, and contains the generic point  $\eta_0$  of  $E_0$ .

(3) By Theorem 80,  $\tilde{C}_m^{(d)} \setminus U^{(d)} = H$ , which by (2) is irreducible of codimension 1 with generic point  $\eta_0$ ; every other point of  $H$  is a proper specialization of  $\eta_0$  and hence has local ring of dimension  $\geq 2$ . By (1),  $\mathcal{O}_{\tilde{C}_m^{(d)}, \eta_0}$  is a regular local ring of dimension 1, i.e. a discrete valuation ring, and it equals  $\mathcal{O}_{X_{m,d}, \eta_0}$  because  $\tilde{C}_m^{(d)}$  is open in  $X_{m,d}$ . Its fraction field is the function field of  $X_{m,d}$ , which equals  $K = k(C^{(d)})$  since  $\pi$  is birational.  $\square$

We denote by  $v_{\eta_0}$  the discrete valuation on  $K$  associated to  $\mathcal{O}_{\tilde{C}_m^{(d)}, \eta_0}$ , and by  $\widehat{K}_{\eta_0}$  the fraction field of the completion.

### Step 1: Purity, and extending unramified torsors

**Proposition 3.** *Let  $G$  be a finite abelian group, let  $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \rightarrow G$  be a continuous homomorphism and let  $\mathcal{Q}$  be the corresponding  $G$ -torsor on  $U_{\acute{e}t}^{(d)}$  (Corollary 14). If  $\mathcal{Q}$  is unramified over  $\tilde{C}_m^{(d)}$  at  $(H, \eta_0)$  in the sense of Definition 54, then  $\rho$  factors, necessarily uniquely, through the canonical surjection*

$$\iota_* : \pi_1^{et}(U^{(d)}, \bar{u}) \twoheadrightarrow \pi_1^{et}(\tilde{C}_m^{(d)}, \bar{u}),$$

where  $\iota : U^{(d)} \hookrightarrow \tilde{C}_m^{(d)}$  is the open immersion. Consequently,  $\mathcal{Q}$  extends to a  $G$ -torsor on  $(\tilde{C}_m^{(d)})_{\acute{e}t}$ , uniquely up to isomorphism.

*Proof. Surjectivity of  $\iota_*$ .* We first note that for every connected finite étale cover  $T \rightarrow \tilde{C}_m^{(d)}$ , the restriction  $T \times_{\tilde{C}_m^{(d)}} U^{(d)}$  is connected:  $T$  is étale over the regular scheme  $\tilde{C}_m^{(d)}$ , hence regular, and being connected it is irreducible;  $T \times_{\tilde{C}_m^{(d)}} U^{(d)}$  is a nonempty open subscheme of  $T$  (nonempty since  $T \rightarrow \tilde{C}_m^{(d)}$  is surjective and  $U^{(d)}$  is dense), hence irreducible, hence connected. This implies surjectivity of  $\iota_*$ : if the image were a proper closed subgroup, it would be contained in a proper open subgroup  $M \leq \pi_1^{et}(\tilde{C}_m^{(d)}, \bar{u})$ , and the connected cover corresponding to  $M$  would pull back to a disconnected cover of  $U^{(d)}$  (the image of  $\pi_1^{et}(U^{(d)})$  does not act transitively on  $\pi_1^{et}(\tilde{C}_m^{(d)})/M$ ), a contradiction.

**The connected cover and its normalization.** Let  $G' = \rho\left(\pi_1^{et}(U^{(d)}, \bar{u})\right) \subseteq G$  and let  $V \rightarrow U^{(d)}$  be the connected finite étale Galois cover with group  $G'$  corresponding to the open normal subgroup  $\ker \rho$  (Theorem 13). As a  $U^{(d)}$ -scheme,  $\mathcal{Q}$  is isomorphic to a disjoint union of  $[G : G']$  copies of  $V$ ; hence at the generic point  $\eta_0$  the finite separable  $\widehat{K}_{\eta_0}$ -algebras of  $\mathcal{Q}$  and of  $V$  have the same field factors, and  $V$  is unramified over  $\tilde{C}_m^{(d)}$  at  $(H, \eta_0)$ .

$V$  is étale over the smooth scheme  $U^{(d)}$ , hence smooth over  $k$ ; being connected it is integral, and  $K(V)/K$  is a finite separable field extension. Let  $\nu : W \rightarrow \tilde{C}_m^{(d)}$  be the normalization of  $\tilde{C}_m^{(d)}$  in  $K(V)$ , i.e. the relative spectrum of the integral closure of  $\mathcal{O}_{\tilde{C}_m^{(d)}}$  in  $K(V)$ . Since  $\tilde{C}_m^{(d)}$  is of finite type over a field it is Nagata ([Stacks, Tag 0335]), so  $\nu$  is a finite morphism, and  $W$  is integral and normal. Over the open  $U^{(d)}$ , the scheme  $V$  is finite over  $U^{(d)}$ , normal, with function field  $K(V)$ ; by uniqueness of normalization,  $\nu^{-1}(U^{(d)}) \cong V$  as  $U^{(d)}$ -schemes.

**$\nu$  is étale, by purity of the branch locus.** We apply [Stacks, Tag 0BMB] at each point  $w \in W$  with image  $y = \nu(w)$ . The hypotheses hold:  $\mathcal{O}_{W,w}$  is normal;  $\mathcal{O}_{\tilde{C}_m^{(d)}, y}$  is regular (Lemma 2);  $\nu$  is finite, hence quasi-finite at  $w$ ; and  $\dim(\mathcal{O}_{W,w}) = \dim(\mathcal{O}_{\tilde{C}_m^{(d)}, y})$ , since for a finite dominant morphism of integral schemes of finite type over a field one has  $\dim(\mathcal{O}_{W,w}) = \dim W - \dim \overline{\{w\}}$ ,  $\dim W = \dim \tilde{C}_m^{(d)}$  and  $\dim \overline{\{w\}} = \dim \overline{\{y\}}$ . It remains to check the codimension-one condition: let  $w' \rightsquigarrow w$  be a specialization with  $\dim(\mathcal{O}_{W,w'}) = 1$ ; then  $y' = \nu(w')$  satisfies  $\dim(\mathcal{O}_{\tilde{C}_m^{(d)}, y'}) = 1$  by the same dimension count. If  $y' \in U^{(d)}$ , then  $\nu$  is étale at  $w'$  since  $\nu^{-1}(U^{(d)}) = V$  is étale over  $U^{(d)}$ . Otherwise  $y' \in \tilde{C}_m^{(d)} \setminus U^{(d)}$ , and by Lemma 2(3) necessarily  $y' = \eta_0$ ; then  $\mathcal{O}_{W,w'}$  is a localization of the integral closure of the discrete valuation ring  $\mathcal{O}_{\tilde{C}_m^{(d)}, \eta_0}$  in  $K(V)$  at one of its maximal ideals, and the hypothesis that  $V$  is unramified at  $(H, \eta_0)$  says precisely (Definition 27) that the ramification index

there is 1 and the residue extension is separable, i.e. that  $\nu$  is unramified at  $w'$ . If  $\dim(\mathcal{O}_{W,w}) = 0$  then  $w$  is the generic point of  $W$  and  $\nu$  is étale at  $w$  because  $K(V)/K$  is finite separable. In all cases [Stacks, Tag 0BMB] applies, and we conclude that  $\nu : W \rightarrow \tilde{C}_m^{(d)}$  is finite étale.

**Factorization.** The geometric point  $\bar{u}$  of  $U^{(d)}$  is also a geometric point of  $\tilde{C}_m^{(d)}$ , and  $W_{\bar{u}} = V_{\bar{u}}$  as finite sets. The action of  $\pi_1^{et}(U^{(d)}, \bar{u})$  on  $V_{\bar{u}}$  is the restriction along  $\iota_*$  of the action of  $\pi_1^{et}(\tilde{C}_m^{(d)}, \bar{u})$  on  $W_{\bar{u}}$ , because  $V = W \times_{\tilde{C}_m^{(d)}} U^{(d)}$ . Since  $V$  is Galois with group  $G'$ , the kernel of the first action is exactly  $\ker \rho$ ; hence  $\ker(\iota_*) \subseteq \ker \rho$ . As  $\iota_*$  is surjective,  $\rho$  factors as  $\rho = \tilde{\rho} \circ \iota_*$  for a unique continuous  $\tilde{\rho} : \pi_1^{et}(\tilde{C}_m^{(d)}, \bar{u}) \rightarrow G' \subseteq G$ . Let  $\tilde{\mathcal{Q}}$  be the  $G$ -torsor on  $(\tilde{C}_m^{(d)})_{\text{ét}}$  corresponding to  $\tilde{\rho}$  (Corollary 14); then  $\iota^* \tilde{\mathcal{Q}} \cong \mathcal{Q}$  since the corresponding characters of  $\pi_1^{et}(U^{(d)}, \bar{u})$  agree. Finally, if  $\tilde{\mathcal{Q}}_1, \tilde{\mathcal{Q}}_2$  are two extensions of  $\mathcal{Q}$ , their characters  $\tilde{\rho}_1, \tilde{\rho}_2$  satisfy  $\tilde{\rho}_1 \circ \iota_* = \tilde{\rho}_2 \circ \iota_*$ , hence are equal by surjectivity of  $\iota_*$ , and  $\tilde{\mathcal{Q}}_1 \cong \tilde{\mathcal{Q}}_2$ .  $\square$

## Step 2: At the boundary, tame implies unramified

The following three lemmas set up the geometry of the generic fiber of  $\tilde{\Phi}_d$ .

**Lemma 4.** *Let  $\xi$  be the generic point of  $\text{Pic}_{C,m}^d$  and  $\kappa := \kappa(\xi)$  its residue field. Write*

$$P_\xi := \tilde{\Phi}_d^{-1}(\xi), \quad U_\xi := \Phi_d^{-1}(\xi) = P_\xi \cap U^{(d)}, \quad H_\xi := H \times_{\text{Pic}_{C,m}^d} \text{Spec}(\kappa),$$

where  $H_\xi$  is a closed subscheme of  $P_\xi$  with underlying set  $P_\xi \setminus U_\xi$ . Then:

1.  $P_\xi \cong \mathbb{P}_\kappa^N$  with  $N = d - \deg \mathfrak{m} - g + 1 \geq 1$ .
2.  $P_\xi$  is a localization of  $\tilde{C}_m^{(d)}$ : for every point  $w \in P_\xi$  one has  $\mathcal{O}_{P_\xi, w} = \mathcal{O}_{\tilde{C}_m^{(d)}, w}$ . In particular the generic point of  $\tilde{C}_m^{(d)}$  lies in  $P_\xi$  and  $\kappa(P_\xi) = K$ .
3.  $\eta_0 \in P_\xi$ , the set  $H_\xi$  is irreducible with generic point  $\eta_0$ , and the reduced closed subscheme  $Z_\xi := (H_\xi)_{\text{red}}$  is a prime divisor on  $P_\xi$  with  $\mathcal{O}_{P_\xi, \eta_0} = \mathcal{O}_{\tilde{C}_m^{(d)}, \eta_0}$ , the discrete valuation ring of  $v_{\eta_0}$ .
4. Every point  $w \neq \eta_0$  of  $P_\xi$  with  $\dim(\mathcal{O}_{P_\xi, w}) = 1$  lies in  $U_\xi$ .
5.  $U_\xi \otimes_\kappa \bar{\kappa} \cong \mathbb{A}_{\bar{\kappa}}^N$ , where  $\bar{\kappa}$  is an algebraic closure of  $\kappa$ .

*Proof.* Throughout we use that  $\text{Pic}_{C,m}^d$  is integral: it is a torsor under the smooth group scheme  $\text{Pic}_{C,m}^0$ , which has geometrically connected fibers, hence it is smooth and geometrically connected over  $k$ , and a connected scheme smooth over a field is integral. Thus  $\kappa = \kappa(\xi)$  is the function field of  $\text{Pic}_{C,m}^d$ .

(1) By Theorem 80,  $\tilde{\Phi}_d$  is a projective space bundle; choosing a trivializing open subset (any nonempty open contains the generic point  $\xi$ ) gives  $P_\xi \cong \mathbb{P}_\kappa^N$  for some  $N$ . Since  $\Phi_d$  is smooth, surjective, of relative dimension  $d - \deg \mathfrak{m} - g + 1$  (Section 4.3) and  $U_\xi = \Phi_d^{-1}(\xi)$  is a nonempty open subscheme of  $P_\xi$ , we get  $N = d - \deg \mathfrak{m} - g + 1$ , and  $N \geq 1$  by (10) (as in the proof of Lemma 2(2)).

(2) Let  $w \in P_\xi$ , choose an affine open  $\text{Spec}(R) \subseteq \text{Pic}_{C,m}^d$  (it contains  $\xi$ , and  $\kappa = \text{Frac}(R)$ ) and an affine open  $\text{Spec}(A) \subseteq \tilde{C}_m^{(d)}$  containing  $w$  with  $\tilde{\Phi}_d(\text{Spec } A) \subseteq \text{Spec } R$ . Then

$$P_\xi \cap \text{Spec}(A) = \text{Spec}(A \otimes_R \text{Frac}(R)) = \text{Spec}\left((R \setminus \{0\})^{-1} A\right)$$

is a localization of  $\text{Spec}(A)$ , whence  $\mathcal{O}_{P_\xi, w} = \mathcal{O}_{\tilde{C}_m^{(d)}, w}$ . For the last assertion, let  $\eta$  be the generic point of the integral scheme  $\tilde{C}_m^{(d)}$  (Lemma 2(1)). Since  $\tilde{\Phi}_d$  is surjective,  $\text{Pic}_{C, m}^d = \tilde{\Phi}_d(\overline{\{\eta\}}) \subseteq \overline{\{\tilde{\Phi}_d(\eta)\}}$ , so  $\tilde{\Phi}_d(\eta) = \xi$  and  $\eta \in P_\xi$ . As  $\mathcal{O}_{P_\xi, \eta} = \mathcal{O}_{\tilde{C}_m^{(d)}, \eta} = K$  is a field,  $\eta$  is the generic point of the integral scheme  $P_\xi$  and  $\kappa(P_\xi) = K$ .

(3) We first claim that  $H \rightarrow \text{Pic}_{C, m}^d$  is surjective. It is proper, since  $H$  is closed in  $\tilde{C}_m^{(d)}$  and  $\tilde{\Phi}_d$  is proper; hence its image is closed. The image contains every closed point  $y$ : otherwise  $\tilde{\Phi}_d^{-1}(y) \subseteq U^{(d)}$ , i.e.  $\Phi_d^{-1}(y) = \tilde{\Phi}_d^{-1}(y)$  would be proper over  $\kappa(y)$ ; but the geometric fibers of  $\Phi_d$  are affine spaces of dimension  $N \geq 1$  (Section 4.3), which are not proper. A closed subset of the Jacobson scheme  $\text{Pic}_{C, m}^d$  containing all closed points is the whole space. Now, since  $H$  is irreducible with generic point  $\eta_0$  (Lemma 2(2)),

$$\text{Pic}_{C, m}^d = \tilde{\Phi}_d(H) = \tilde{\Phi}_d(\overline{\{\eta_0\}}) \subseteq \overline{\{\tilde{\Phi}_d(\eta_0)\}},$$

so  $\tilde{\Phi}_d(\eta_0) = \xi$  and  $\eta_0 \in H_\xi$ .

For irreducibility of  $H_\xi$  we may replace  $H$  by  $H_{red}$ , an integral scheme dominating  $\text{Pic}_{C, m}^d$  (this does not change the underlying topological space of  $H_\xi$ ). As in (2),  $H_\xi$  is covered by the open subschemes  $\text{Spec}((R \setminus \{0\})^{-1}A)$ , where  $\text{Spec}(A) \subseteq H_{red}$  runs over affine opens mapping into affine opens  $\text{Spec}(R) \subseteq \text{Pic}_{C, m}^d$ . Each such  $A$  is a domain and, since the nonempty open  $\text{Spec}(A)$  of the irreducible  $H_{red}$  dominates  $\text{Pic}_{C, m}^d$ , the map  $R \rightarrow A$  is injective; hence  $(R \setminus \{0\})^{-1}A$  is a nonzero localization of a domain, and each nonempty piece is irreducible with generic point the generic point  $\eta_0$  of  $H_{red}$ . A topological space covered by irreducible opens all containing one common point is irreducible; thus  $H_\xi$  is irreducible with generic point  $\eta_0$ . Finally, by (2) and Lemma 2(3),  $\mathcal{O}_{P_\xi, \eta_0} = \mathcal{O}_{\tilde{C}_m^{(d)}, \eta_0}$  is a discrete valuation ring, so  $Z_\xi = \overline{\{\eta_0\}} \subseteq P_\xi$  (with its reduced structure) is a prime divisor on  $P_\xi \cong \mathbb{P}_\kappa^N$ .

(4) Let  $w \in P_\xi$  with  $\dim(\mathcal{O}_{P_\xi, w}) = 1$  and  $w \neq \eta_0$ . By (2),  $\dim(\mathcal{O}_{\tilde{C}_m^{(d)}, w}) = 1$ . If  $w \notin U^{(d)}$  then, by Lemma 2(3),  $w = \eta_0$ , a contradiction. Hence  $w \in U^{(d)} \cap P_\xi = U_\xi$ .

(5) This is the description of the geometric fibers of  $\Phi_d$  recalled in Section 4.3, applied to the geometric point  $\text{Spec}(\bar{\kappa}) \rightarrow \text{Pic}_{C, m}^d$  lying over  $\xi$ .  $\square$

The next two lemmas identify the divisor  $Z_\xi$  inside  $P_\xi \cong \mathbb{P}_\kappa^N$ . Note that  $\kappa$  is in general neither perfect nor separably closed, so  $Z_\xi$  need not be a hyperplane over  $\kappa$ ; the lemmas show that it is a hyperplane geometrically, and that the inseparability phenomena can only affect its degree by a factor which is a power of  $p$  — which is all we shall need.

**Lemma 5.** *Let  $L$  be an algebraically closed field, let  $N \geq 1$ , and let  $T \subsetneq \mathbb{P}_L^N$  be a closed subset such that  $\mathbb{P}_L^N \setminus T \cong \mathbb{A}_L^N$  as  $L$ -schemes. Then  $T$  is a hyperplane.*

*Proof.* Write  $V := \mathbb{P}_L^N \setminus T$  and fix an isomorphism  $V \cong \mathbb{A}_L^N$ , so  $\mathcal{O}(V) \cong L[t_1, \dots, t_N]$  and  $\mathcal{O}(V)^\times = L^\times$ . Let  $T_1, \dots, T_r$  be the irreducible components of  $T$  of codimension 1 in  $\mathbb{P}_L^N$ ; each is an integral hypersurface  $T_i = V_+(f_i)$  with  $f_i$  an irreducible form of degree  $d_i \geq 1$ .

*There is at least one component:*  $r \geq 1$ . Otherwise  $T$  has codimension  $\geq 2$ , and any  $h \in \mathcal{O}(V)$ , viewed as a rational function on  $\mathbb{P}_L^N$ , has polar locus contained in  $T$ , hence of codimension  $\geq 2$ ; since  $\mathbb{P}_L^N$  is normal,  $h$  is then regular on all of  $\mathbb{P}_L^N$ , hence constant. This contradicts  $\mathcal{O}(V) \cong L[t_1, \dots, t_N]$ .

*There is exactly one:*  $r = 1$ . If  $r \geq 2$ , then  $h := f_1^{d_2}/f_2^{d_1}$  is a well-defined rational function on  $\mathbb{P}_L^N$  (numerator and denominator are forms of the same degree  $d_1 d_2$ ), regular and invertible outside

$T_1 \cup T_2 \subseteq T$ . Thus  $h|_V \in \mathcal{O}(V)^\times = L^\times$ , so  $h$  is constant and  $0 = \operatorname{div}(h) = d_2 T_1 - d_1 T_2$ , which is absurd as  $T_1 \neq T_2$  are distinct prime divisors.

*T has no components of codimension  $\geq 2$ :*  $T = T_1$ . Let  $V_1 := \mathbb{P}_L^N \setminus T_1 \supseteq V$ , an affine open (the complement of a hypersurface in projective space), with  $V_1 \setminus V = T \setminus T_1$  closed in  $V_1$  of codimension  $\geq 2$ . For  $h \in \mathcal{O}(V)$ , the polar locus of  $h$  on the normal scheme  $V_1$  is either empty or of pure codimension one, and it is contained in  $V_1 \setminus V$ ; hence it is empty and  $h \in \mathcal{O}(V_1)$ . So the restriction  $\mathcal{O}(V_1) \rightarrow \mathcal{O}(V)$  is bijective (injectivity holds as  $V$  is dense), and the open immersion  $V \hookrightarrow V_1$  of affine schemes induces an isomorphism on coordinate rings, hence is an isomorphism. Therefore  $T = T_1$ .

$d_1 = 1$ . Choose any hyperplane  $W \subset \mathbb{P}_L^N$  with  $W \neq T_1$ . Then  $W \cap V \neq \emptyset$  (otherwise  $W \subseteq T_1$ , and both being integral of the same dimension,  $W = T_1$ ), and  $W \cap V$  is a prime divisor on  $V \cong \mathbb{A}_L^N$ . As  $L[t_1, \dots, t_N]$  is a unique factorization domain, the corresponding height-one prime is principal, generated by an irreducible polynomial  $h$ , and  $\operatorname{div}_V(h) = [W \cap V]$ . Viewing  $h$  as a rational function on  $\mathbb{P}_L^N$ , all components of  $\operatorname{div}_{\mathbb{P}_L^N}(h)$  other than  $\overline{W \cap V} = W$  are supported on  $T$ , whose unique codimension-one component is  $T_1$ ; hence  $\operatorname{div}_{\mathbb{P}_L^N}(h) = [W] + a[T_1]$  for some  $a \in \mathbb{Z}$ . Taking degrees,  $0 = 1 + a d_1$ , so  $d_1 \mid 1$ , i.e.  $d_1 = 1$ .

Thus  $T = T_1 = V_+(f_1)$  with  $f_1$  of degree one: a hyperplane.  $\square$

**Lemma 6.** *Let  $\kappa$  be a field with algebraic closure  $\bar{\kappa}$ , and let  $Z \subset \mathbb{P}_{\bar{\kappa}}^N$  be an integral hypersurface of degree  $\delta$  such that the support of  $Z \otimes_{\bar{\kappa}} \bar{\kappa}$  is a single hyperplane of  $\mathbb{P}_{\bar{\kappa}}^N$ . If  $\operatorname{char}(\kappa) = 0$  then  $\delta = 1$ ; if  $\operatorname{char}(\kappa) = p > 0$  then  $\delta$  is a power of  $p$ .*

*Proof.* Let  $g \in \kappa[x_0, \dots, x_N]$  be an irreducible form defining  $Z$ , so  $\deg g = \delta$ , and let  $\ell$  be a linear form over  $\bar{\kappa}$  defining the hyperplane  $\operatorname{supp}(Z_{\bar{\kappa}})$ .

*Step 1:*  $g = c \ell^\delta$  in  $\bar{\kappa}[x_0, \dots, x_N]$  for some  $c \in \bar{\kappa}^\times$ . Let  $q$  be any irreducible factor of  $g$  over  $\bar{\kappa}$ . Then  $V_+(q)$  is an integral hypersurface contained in  $\operatorname{supp}(Z_{\bar{\kappa}}) = V_+(\ell)$ ; both are irreducible closed subsets of dimension  $N - 1$ , hence  $V_+(q) = V_+(\ell)$ . Since  $(q)$  and  $(\ell)$  are prime ideals with the same zero set, they have the same radical, hence are equal, and  $q$  is a scalar multiple of  $\ell$ . As this holds for every irreducible factor,  $g = c \ell^\delta$ .

*Step 2:*  $g$  remains irreducible over the separable closure  $\kappa^{\operatorname{sep}}$ . First,  $g$  is squarefree in  $\kappa^{\operatorname{sep}}[x_0, \dots, x_N]$ . Indeed, suppose  $q^2 \mid g$  for some irreducible  $q \in \kappa^{\operatorname{sep}}[x_0, \dots, x_N]$  of positive degree. The coefficients of  $q$  lie in a finite Galois extension  $\kappa'/\kappa$ ; normalize  $q$  so that its leading coefficient (in some fixed monomial order) is 1, and let  $q = q_1, \dots, q_t$  be the distinct  $\operatorname{Gal}(\kappa'/\kappa)$ -conjugates of  $q$ , normalized the same way. Applying elements of  $\operatorname{Gal}(\kappa'/\kappa)$  to the relation  $q^2 \mid g$  shows  $q_j^2 \mid g$  for all  $j$ ; the  $q_j$  are pairwise coprime irreducibles, so  $M^2 \mid g$  in  $\kappa'[x_0, \dots, x_N]$  where  $M := q_1 \cdots q_t$ . The polynomial  $M$  is fixed by  $\operatorname{Gal}(\kappa'/\kappa)$  (the Galois action permutes the normalized  $q_j$ ), so  $M \in \kappa[x_0, \dots, x_N]$ . Writing  $g = M^2 h$  with  $h \in \kappa'[x_0, \dots, x_N]$ , and expanding  $h$  in a  $\kappa$ -basis of  $\kappa'$ , the relation  $g, M \in \kappa[x_0, \dots, x_N]$  forces  $h \in \kappa[x_0, \dots, x_N]$  (clear denominators: if  $Dh \in \kappa[x_0, \dots, x_N]$  for some nonzero  $D \in \kappa[x_0, \dots, x_N]$ , then all coordinates of  $h$  outside  $\kappa$  vanish). This contradicts the irreducibility of  $g$  over  $\kappa$ . Hence  $g$  is squarefree over  $\kappa^{\operatorname{sep}}$ , say  $g = c' q_1 \cdots q_t$  with  $q_i \in \kappa^{\operatorname{sep}}[x_0, \dots, x_N]$  pairwise non-associate irreducibles. Second,  $t = 1$ : for each  $i$ , the closed subset  $V_+(q_i)$  over  $\bar{\kappa}$  is contained in  $V_+(\ell)$  and has dimension  $N - 1$ , hence equals  $V_+(\ell)$ ; since the projection  $\mathbb{P}_{\bar{\kappa}}^N \rightarrow \mathbb{P}_{\kappa^{\operatorname{sep}}}^N$  is surjective and maps  $V_+(q_i)$  onto  $V_+(q_i)$ , we get  $V_+(q_1) = V_+(q_2) = \dots$  in  $\mathbb{P}_{\kappa^{\operatorname{sep}}}^N$ , so (equality of radicals of primes, as in Step 1) the  $q_i$  are pairwise associate — possible only if  $t = 1$ . Thus  $g$  is irreducible over  $\kappa^{\operatorname{sep}}$ , and we may and do replace  $\kappa$  by  $\kappa^{\operatorname{sep}}$ .



*Step 3: conclusion.* If  $\text{char}(\kappa) = 0$ , then  $\kappa^{sep} = \bar{\kappa}$  and  $g = c\ell^\delta$  is irreducible, forcing  $\delta = 1$ . Assume  $\text{char}(\kappa) = p > 0$  and  $\kappa$  separably closed, so  $\bar{\kappa}/\kappa$  is purely inseparable. Write  $\ell = \sum_i \lambda_i x_i$  with  $\lambda_i \in \bar{\kappa}$ , and choose  $r \geq 0$  with  $\lambda_i^{p^r} \in \kappa$  for all  $i$ . Then

$$q := \ell^{p^r} = \sum_i \lambda_i^{p^r} x_i^{p^r} \in \kappa[x_0, \dots, x_N],$$

and  $q^\delta = \ell^{p^r \delta} = c^{-p^r} g^{p^r}$ . Both  $q^\delta$  and  $g^{p^r}$  lie in  $\kappa[x_0, \dots, x_N]$ , so comparing any monomial appearing with nonzero coefficient shows  $c^{-p^r} \in \kappa^\times$ . Now in the unique factorization domain  $\kappa[x_0, \dots, x_N]$ , every irreducible factor of  $q$  divides  $g^{p^r}$ , hence is an associate of the irreducible  $g$ ; therefore  $q = c'' g^s$  for some  $s \geq 1$  and  $c'' \in \kappa^\times$ . Comparing degrees,  $p^r = s\delta$ , so  $\delta$  divides  $p^r$  and is a power of  $p$ .  $\square$

We can now prove the key proposition.

**Proposition 7.** *Let  $G$  be a finite abelian group and let  $\mathcal{Q}$  be a  $G$ -torsor on  $U_{\text{ét}}^{(d)}$ . If  $\mathcal{Q}$  is tamely ramified over  $\tilde{C}_{\mathfrak{m}}^{(d)}$  at  $(H, \eta_0)$ , then  $\mathcal{Q}$  is unramified at  $(H, \eta_0)$ .*

Note that no bound on the ramification of  $\mathcal{Q}$  along  $\mathfrak{m}$  is assumed; being a torsor on all of  $U^{(d)}$  is enough.

*Proof.* As in Section 3 (see the discussion following Definition 55), the restriction of  $\mathcal{Q}$  to  $\text{Spec}(\hat{K}_{\eta_0})$  decomposes as a disjoint union of copies of  $\text{Spec}(M)$  for a finite Galois extension  $M/\hat{K}_{\eta_0}$  with Galois group a subgroup of  $G$ , and corresponds to a continuous homomorphism

$$\rho : G_{\hat{K}_{\eta_0}} \rightarrow G$$

with image  $\text{Gal}(M/\hat{K}_{\eta_0})$ . Let  $I \subseteq G_{\hat{K}_{\eta_0}}$  be the inertia subgroup. Then  $\mathcal{Q}$  is unramified at  $(H, \eta_0)$  if and only if  $\rho(I) = \{1\}$ , and  $\rho(I)$  equals the inertia subgroup  $G_0$  of  $\text{Gal}(M/\hat{K}_{\eta_0})$ .

**Reduction to  $k$  algebraically closed.** Let  $k'/k$  be a finite separable extension. The projection  $\text{pr} : (\tilde{C}_{\mathfrak{m}}^{(d)})_{k'} \rightarrow \tilde{C}_{\mathfrak{m}}^{(d)}$  is (finite) étale, and it identifies  $(\tilde{C}_{\mathfrak{m}}^{(d)})_{k'} \setminus (U^{(d)})_{k'}$  with  $\text{pr}^{-1}(H)$ . Since  $C^{(n)}$  is geometrically integral for every  $n$  (see the Preliminaries), so are  $Z_0$ ,  $C^{(d)}$  and hence  $E_0$  (the exceptional divisor of a blowup of a geometrically integral smooth scheme along a geometrically integral smooth center); therefore  $\text{pr}^{-1}(H)$  is irreducible with a unique generic point  $\eta'$  lying over  $\eta_0$ , and  $\text{pr}$  is étale at  $\eta'$ . By Lemma 59 (and its proof: the completed local extensions at  $\eta'$  and at  $\eta_0$  differ by the unramified base change  $\hat{K}_{\eta'}/\hat{K}_{\eta_0}$ ), the torsor  $\text{pr}^{-1}\mathcal{Q}$  is tamely ramified, resp. unramified, at  $\eta'$  if and only if  $\mathcal{Q}$  is tamely ramified, resp. unramified, at  $\eta_0$ . Passing to the colimit over all finite separable  $k'/k$  (recall  $k$  is perfect, so  $\bar{k} = k^{sep}$ ), we may and do assume for the rest of the proof that  $k$  is algebraically closed.

**The inertia image is cyclic of order prime to  $p$ .** Set  $G_0 := \rho(I)$ , the inertia group of  $M/\hat{K}_{\eta_0}$ . Since  $\mathcal{Q}$  is tamely ramified at  $(H, \eta_0)$ , the extension  $M/\hat{K}_{\eta_0}$  is tamely ramified with respect to  $\mathcal{O}_{\tilde{C}_{\mathfrak{m}}^{(d)}, \eta_0}$  (Definition 27): its residue extension is separable and its ramification index  $e$  is prime to  $p$ . For a Galois extension of complete discrete valuation rings with separable residue extension, the inertia group  $G_0$  has order equal to the ramification index ([Ser79], Ch. IV); hence  $|G_0| = e$  is prime to  $p$ . Furthermore, the wild inertia subgroup  $G_1 \subseteq G_0$  is a  $p$ -group and  $G_0/G_1$  is cyclic ([Ser79], Ch. IV; recalled in the Preliminaries); since  $|G_0|$  is prime to  $p$  this forces  $G_1 = \{1\}$ , so  $G_0$  is cyclic of order  $e$  prime to  $p$ .

**Choice of a character.** Write  $G = G_p \times G'$  with  $G_p$  a  $p$ -group and  $|G'|$  prime to  $p$ . Since  $\gcd(|G_0|, |G_p|) = 1$ , the projection  $\text{pr}_{G'} : G \rightarrow G'$  is injective on  $G_0$ . The image  $\text{pr}_{G'}(G_0)$  is cyclic,

so it admits a faithful character  $\mathrm{pr}_{G'}(G_0) \rightarrow \mathbb{Q}/\mathbb{Z}$ , which extends to a character  $\chi_0 : G' \rightarrow \mathbb{Q}/\mathbb{Z}$  because  $\mathbb{Q}/\mathbb{Z}$  is a divisible, hence injective, abelian group. Set  $\chi := \chi_0 \circ \mathrm{pr}_{G'} : G \rightarrow \mathbb{Q}/\mathbb{Z}$  and let  $m := |\chi(G)|$ , so that  $\chi(G) = \frac{1}{m}\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/m$ . Then  $m$  divides  $|G'|$ , hence is prime to  $p$ , and  $\chi$  is injective on  $G_0$ . Since  $k = \bar{k}$  and  $p \nmid m$ , we may fix an isomorphism  $\mathbb{Z}/m \cong \mu_m(k)$  and regard  $\chi$  as a surjection  $\chi : G \rightarrow \mu_m$ .

Let  $\mathcal{Q}_\chi := \mathcal{Q} \times^G \mu_m$  be the  $\mu_m$ -torsor on  $U_{\mathrm{ét}}^{(d)}$  obtained from  $\mathcal{Q}$  by pushing forward along  $\chi$  (Proposition 15, applied to the quotient  $G \rightarrow G/\ker \chi$  followed by the isomorphism  $G/\ker \chi \cong \mu_m$  induced by  $\chi$ ). By functoriality of the correspondence of Corollary 14, the restriction of  $\mathcal{Q}_\chi$  to  $\mathrm{Spec}(\hat{K}_{\eta_0})$  corresponds to the character  $\chi \circ \rho$ . It therefore suffices to prove that  $\mathcal{Q}_\chi$  is unramified at  $(H, \eta_0)$ : this gives  $\chi(\rho(I)) = \{1\}$ , and since  $\chi$  is injective on  $G_0 = \rho(I)$  we conclude  $\rho(I) = \{1\}$ , as desired.

**Kummer representative.** Restrict the character of  $\mathcal{Q}_\chi$  to the absolute Galois group of  $K = k(U^{(d)}) = k(C^{(d)})$ . Since  $\mu_m \subset k \subseteq K$ , Kummer theory identifies  $H^1(K, \mu_m) \cong K^\times / (K^\times)^m$ ; let  $F \in K^\times$  be a representative of the resulting class, so that the corresponding  $\mu_m$ -cover of  $\mathrm{Spec}(K)$  is  $\mathrm{Spec}(K[y]/(y^m - F))$ , and  $F$  is well defined up to  $m$ -th powers. We will show:

$$v_{\eta_0}(F) \equiv 0 \pmod{m}. \quad (1)$$

Let us first explain why (1) finishes the proof. Let  $\varpi$  be a uniformizer of  $B_0 := \mathcal{O}_{\hat{C}_m^{(d)}, \eta_0}$  and write  $v_{\eta_0}(F) = ma$ ; replacing  $F$  by  $F\varpi^{-ma}$  (this changes neither the class in  $H^1(\hat{K}_{\eta_0}, \mu_m)$  nor the extension, since it changes  $F$  by the  $m$ -th power  $(\varpi^{-a})^m$ ), we may assume  $F \in \hat{B}_0^\times$ , where  $\hat{B}_0$  is the completion. Then

$$B := \hat{B}_0[y]/(y^m - F)$$

is finite étale over  $\hat{B}_0$ : it is finite free, and the derivative  $my^{m-1}$  is a unit in  $B$  because  $m$  is invertible in  $k$  and  $y^m = F$  is a unit. Consequently  $\hat{K}_{\eta_0}[y]/(y^m - F) = B \otimes_{\hat{B}_0} \hat{K}_{\eta_0}$  is a product of finite separable field extensions of  $\hat{K}_{\eta_0}$ , each unramified with respect to  $\hat{B}_0$  (Definition 27; compare Theorem 30(1),(2), whose proof for  $m$  prime to  $p$  does not use perfectness of the residue field). Thus  $\mathcal{Q}_\chi$  is unramified at  $(H, \eta_0)$ , as desired.

**The divisor of  $F$  on the generic fiber.** It remains to establish (1). We use the notation of Lemma 4. By part (2) of that lemma,  $F$  is a nonzero rational function on  $P_\xi \cong \mathbb{P}_\kappa^N$ ; write

$$\mathrm{div}_{P_\xi}(F) = v_{\eta_0}(F) \cdot Z_\xi + \sum_j a_j W_j,$$

where the  $W_j$  are the prime divisors of  $P_\xi$  other than  $Z_\xi$  occurring in  $\mathrm{div}_{P_\xi}(F)$ , and the coefficient of  $Z_\xi$  is indeed  $v_{\eta_0}(F)$  because  $\mathcal{O}_{P_\xi, \eta_0} = \mathcal{O}_{\hat{C}_m^{(d)}, \eta_0}$  (Lemma 4(3)).

All the  $a_j$  are divisible by  $m$ . Let  $w_j$  be the generic point of  $W_j$ . By Lemma 4(4),  $w_j \in U_\xi \subseteq U^{(d)}$ , and  $A_j := \mathcal{O}_{P_\xi, w_j} = \mathcal{O}_{\hat{C}_m^{(d)}, w_j}$  is a discrete valuation ring with fraction field  $K$ . The torsor  $\mathcal{Q}_\chi$  is a torsor on all of  $U_{\mathrm{ét}}^{(d)}$ , hence is finite étale over a neighborhood of  $w_j$ , so it is unramified at  $(W_j, w_j)$  (in the sense of Definition 54, with ambient scheme  $P_\xi$ ). On the other hand, if  $m \nmid a_j = v_{w_j}(F)$ , then in any extension of the valuation  $v_{w_j}$  to the algebra  $K[y]/(y^m - F)$  one has  $v(y) = v_{w_j}(F)/m \notin v_{w_j}(K^\times)$ , so the ramification index over  $A_j$  is  $m/\mathrm{gcd}(m, a_j) > 1$  — contradicting unramifiedness. Hence  $m \mid a_j$  for all  $j$ .

The degree of  $Z_\xi$  is prime to  $m$ . By Lemma 4(5) we have  $U_\xi \otimes_\kappa \bar{\kappa} \cong \mathbb{A}_{\bar{\kappa}}^N$ , and the complement of this open subset in  $P_\xi \otimes_\kappa \bar{\kappa} \cong \mathbb{P}_{\bar{\kappa}}^N$  is exactly the support of  $Z_\xi \otimes_\kappa \bar{\kappa}$  (taking supports and complements



commutes with the base change  $\bar{\kappa}/\kappa$ . By Lemma 5, this support is a single hyperplane of  $\mathbb{P}_{\bar{\kappa}}^N$ . Therefore Lemma 6 applies to the integral hypersurface  $Z_{\xi} \subset \mathbb{P}_{\kappa}^N$ : its degree  $\delta := \deg Z_{\xi}$  equals 1 if  $p = 0$ , and is a power of  $p$  if  $p > 0$ . In either case  $\gcd(\delta, m) = 1$ , since  $m$  is prime to  $p$ .

*Conclusion.* A principal divisor on  $\mathbb{P}_{\kappa}^N$  has degree zero (the numerator and denominator of a rational function are forms of equal degrees). Taking degrees in the displayed formula above,

$$0 = v_{\eta_0}(F) \cdot \delta + \sum_j a_j \deg W_j \equiv v_{\eta_0}(F) \cdot \delta \pmod{m}.$$

Since  $\delta$  is invertible modulo  $m$ , this gives  $v_{\eta_0}(F) \equiv 0 \pmod{m}$ , which is (1). As explained above, this proves that  $\mathcal{Q}_{\chi}$  is unramified at  $(H, \eta_0)$ , hence  $\rho(I) = \{1\}$  and  $\mathcal{Q}$  is unramified at  $(H, \eta_0)$ .  $\square$

## Conclusion of the proof, and the fundamental-group formulation

*Proof of Lemma 1.* Let  $G = \Lambda^{\times}$  and let  $\mathcal{P}^{[d]}$  be the  $G$ -torsor on  $U_{\text{ét}}^{(d)}$  corresponding to  $\mathcal{F}^{[d]}$  under Proposition 9, with character  $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \rightarrow G$ . By Lemma 2(3), the only codimension-one point of  $\tilde{C}_{\mathfrak{m}}^{(d)} \setminus U^{(d)}$  is  $\eta_0$ , so the hypothesis that  $\mathcal{F}^{[d]}$  is tamely ramified along  $H$  means precisely that  $\mathcal{P}^{[d]}$  is tamely ramified over  $\tilde{C}_{\mathfrak{m}}^{(d)}$  at  $(H, \eta_0)$  (Definition 54). By Proposition 7,  $\mathcal{P}^{[d]}$  is then unramified at  $(H, \eta_0)$ . By Proposition 3, the character  $\rho$  factors uniquely as  $\rho = \tilde{\rho} \circ \iota_*$  through  $\pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)}, \bar{u})$ . Let  $\tilde{\mathcal{F}}^{[d]}$  be the locally constant sheaf of free rank-one  $\Lambda$ -modules on  $(\tilde{C}_{\mathfrak{m}}^{(d)})_{\text{ét}}$  corresponding to  $\tilde{\rho}$  (Corollary 14 together with Proposition 9). Since  $\tilde{\rho} \circ \iota_* = \rho$ , its restriction to  $U^{(d)}$  is isomorphic to  $\mathcal{F}^{[d]}$ .  $\square$

*Remark.* The proof also gives uniqueness: since  $\iota_* : \pi_1^{et}(U^{(d)}) \rightarrow \pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)})$  is surjective (Proposition 3), the extension  $\tilde{\mathcal{F}}^{[d]}$  is unique up to isomorphism, and more generally the restriction functor from locally constant sheaves on  $\tilde{C}_{\mathfrak{m}}^{(d)}$  to locally constant sheaves on  $U^{(d)}$  is fully faithful. This is the source of the uniqueness assertions in Theorem 87 and Theorem 2.

We conclude with the promised fundamental-group formulation.

**Corollary 8.** *Let  $G$  be a finite abelian group, and call a continuous homomorphism  $\rho : \pi_1^{et}(U^{(d)}, \bar{u}) \rightarrow G$  tame if the associated  $G$ -torsor on  $U_{\text{ét}}^{(d)}$  (Corollary 14) is tamely ramified over  $\tilde{C}_{\mathfrak{m}}^{(d)}$  at  $(H, \eta_0)$ . Then composition with the canonical homomorphism*

$$\pi_1^{et}(U^{(d)}, \bar{u}) \longrightarrow \pi_1^{et}(\text{Pic}_{C, \mathfrak{m}}^d, \tilde{\Phi}_d(\bar{u})) \quad (\text{induced by } \Phi_d)$$

*is a bijection*

$$\text{Hom}_{\text{cont}}\left(\pi_1^{et}(\text{Pic}_{C, \mathfrak{m}}^d), G\right) \xrightarrow{\cong} \left\{ \rho \in \text{Hom}_{\text{cont}}\left(\pi_1^{et}(U^{(d)}), G\right) : \rho \text{ is tame} \right\}.$$

*Proof.* Since  $\tilde{\Phi}_d$  is proper and smooth with geometrically connected fibers (isomorphic to projective spaces), and projective spaces over separably closed fields are simply connected ([Ten15] Example 4.9, [Töt11] Example 1.4.12), the exact sequence of Corollary 78 shows that  $(\tilde{\Phi}_d)_* : \pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)}) \rightarrow \pi_1^{et}(\text{Pic}_{C, \mathfrak{m}}^d)$  is an isomorphism. Moreover  $\iota_* : \pi_1^{et}(U^{(d)}) \rightarrow \pi_1^{et}(\tilde{C}_{\mathfrak{m}}^{(d)})$  is surjective (Proposition 3), and the homomorphism in the statement is  $(\tilde{\Phi}_d)_* \circ \iota_*$ .

*The map lands in the tame classes, i.e. is well defined.* If  $\rho = \tilde{\rho} \circ (\tilde{\Phi}_d)_* \circ \iota_*$ , the  $G$ -torsor of  $\rho$  is the restriction to  $U^{(d)}$  of a  $G$ -torsor  $\tilde{\mathcal{Q}}$  on  $(\tilde{C}_{\mathfrak{m}}^{(d)})_{\text{ét}}$ . As a scheme,  $\tilde{\mathcal{Q}}$  is finite étale over  $\tilde{C}_{\mathfrak{m}}^{(d)}$ , so

its restriction to  $\text{Spec}(\widehat{K}_{\eta_0})$  is the generic fiber of a finite étale algebra over the completed discrete valuation ring, hence a product of unramified extensions of  $\widehat{K}_{\eta_0}$ . Thus the torsor of  $\rho$  is unramified, in particular tamely ramified, at  $(H, \eta_0)$ .

*Injectivity* is immediate from the surjectivity of  $(\tilde{\Phi}_d)_* \circ \iota_*$ .

*Surjectivity.* Let  $\rho$  be tame. By [Proposition 7](#) its torsor is unramified at  $(H, \eta_0)$ , and by [Proposition 3](#)  $\rho$  factors through  $\iota_*$ ; composing with the inverse of  $(\tilde{\Phi}_d)_*$  exhibits  $\rho$  in the image.  $\square$

*Remark.* [Corollary 8](#) is the precise form of “Route 2”: the canonical surjection from the maximal “tame-along- $H$ ” abelian quotient of  $\pi_1^{et}(U^{(d)})$  to  $\pi_1^{et,ab}(\text{Pic}_{C,m}^d)$  is an isomorphism — not merely injective onto its image. It also implies the statement of “Route 1”: the kernel of  $\pi_1^{ab}(U^{(d)}) \rightarrow \pi_1^{ab}(\text{Pic}_{C,m}^d)$  is a pro- $p$  group. Indeed, let  $K$  denote this kernel and suppose it had a nontrivial finite quotient of order divisible by a prime  $q \neq p$ ; then  $K$  admits a continuous character of order  $q$ , which extends to a continuous character  $\chi$  of  $\pi_1^{ab}(U^{(d)})$  of some finite order  $p^b n$  with  $p \nmid n$ , and  $\chi^{p^b}$  is a character of order prime to  $p$  that is still nontrivial on  $K$ . But a character of order  $n$  prime to  $p$  is automatically tame at  $(H, \eta_0)$ : the corresponding local extensions of  $\widehat{K}_{\eta_0}$  have degree dividing  $n$ , so their ramification indices and residue degrees are prime to  $p$ , making the residue extensions separable and the ramification tame. By [Corollary 8](#),  $\chi^{p^b}$  then factors through  $\pi_1^{ab}(\text{Pic}_{C,m}^d)$ , i.e. vanishes on  $K$  — a contradiction. Hence every finite quotient of  $K$  is a  $p$ -group.

*Remark.* Statements of this kind appear in both of the works our approach follows, with different proofs. Guignard ([\[Gui19\]](#), Lemmas 5.7 and 5.9) does not extend  $\mathcal{F}^{[d]}$  over a compactification: he shows that  $\mathcal{F}^{[d]}$  is constant on the (affine space) fibers of  $\Phi_d$ , using the triviality of the tame fundamental group of  $\mathbb{A}^1$ , and then descends it along  $\Phi_d$  directly by a duality argument. Takeuchi ([\[Tak19\]](#), proof of Theorem 1.2 in §4) obtains the extension of the corresponding character over  $\tilde{C}_m^{(d)}$  from his refined (Kato–Matsuda) Swan conductor computation ([\[Tak19\]](#), Theorem 2.7 and Corollary 2.8), which for  $p$ -power order characters shows unramifiedness at  $\eta_0$  directly. The proof given here is elementary: beyond the purity of the branch locus, it only uses Kummer theory and the geometry of the generic fiber of the compactified Abel–Jacobi fibration — the boundary of an affine space inside projective space is geometrically a hyperplane, and a principal divisor on projective space has degree zero. It is in this sense the formal counterpart of Guignard’s approach.

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